

CS & IT ENGINEERING



Data Structure & Programming

Practice Session

Lecture No.-02



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Topics to be Covered



Practice Session





Topic : PYQ Practice – 1 – GATE 2007 – Homework Question

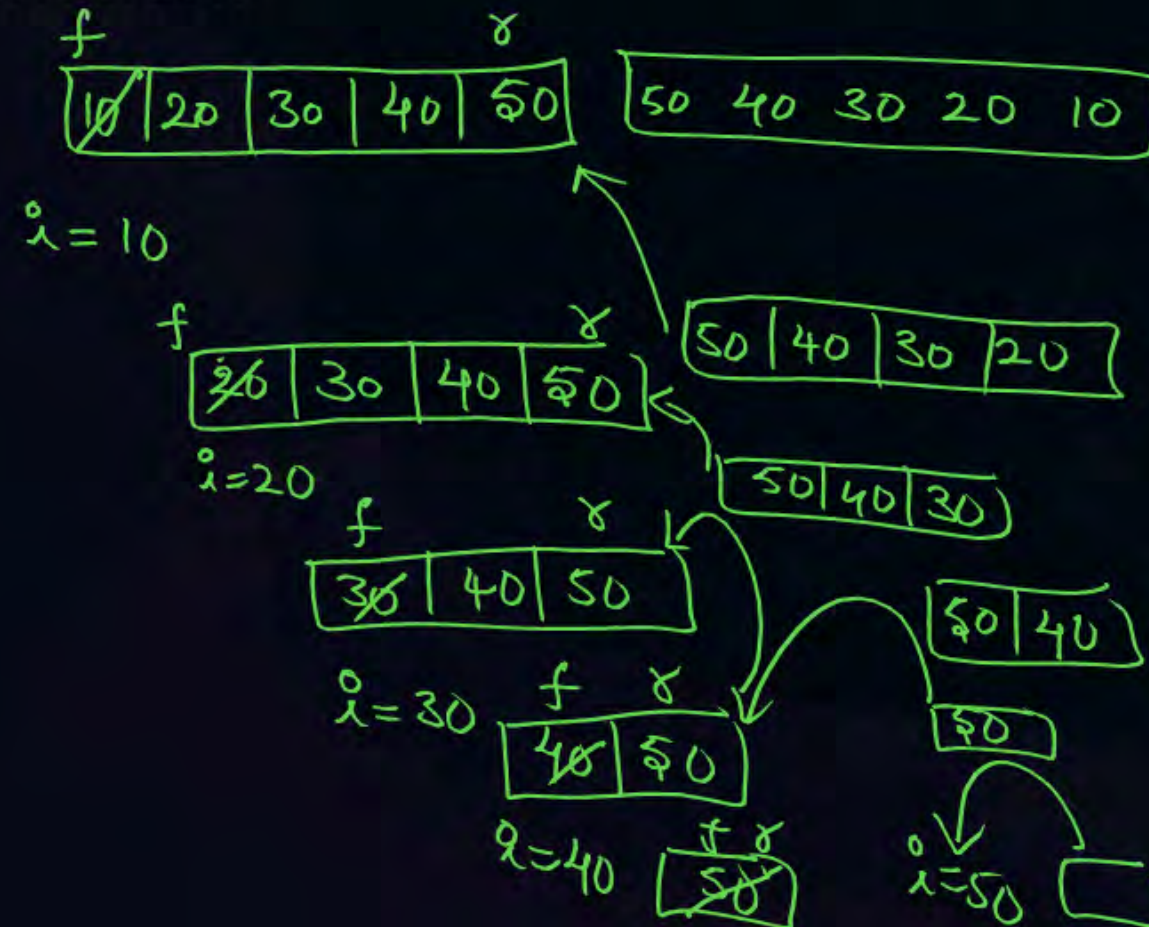


#Q. Suppose you are given an implementation of a queue of integers. The operations that can be performed on the queue are:

- isEmpty (Q) — returns true if the queue is empty, false otherwise.
- delete (Q) — deletes the element at the front of the queue and returns its value.
- insert (Q, i) — inserts the integer i at the rear of the queue.

Consider the following function:

```
void f (queue Q) {  
    int i ;  
    if (!isEmpty(Q)) {  
        i = delete(Q);  
        f(Q);  
        insert(Q, i);  
    }  
}
```





Topic : PYQ Practice – 1 – Homework Question



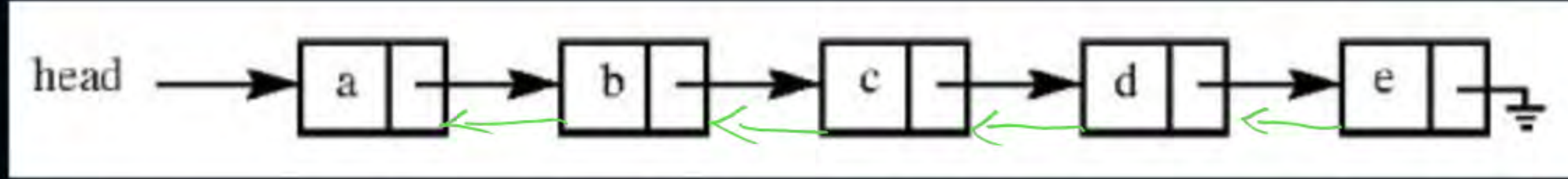
What operation is performed by the above function f ?

- (a) Leaves the queue Q unchanged
- (b) Reverses the order of the elements in the queue Q
- (c) Deletes the element at the front of the queue Q and inserts it at the rear keeping the other elements in the same order
- (d) Empties the queue Q

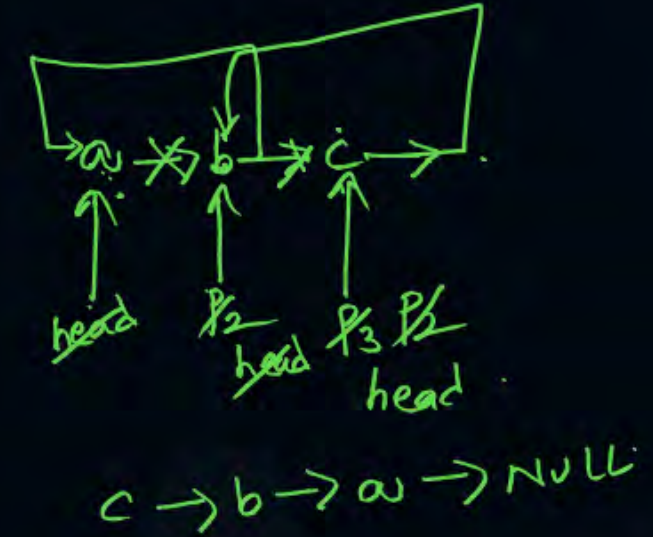
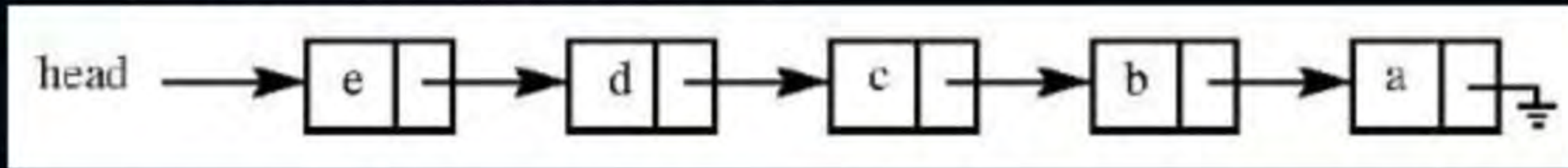


Topic : PYQ Practice - 1 – Homework Question

#Q. Consider the problem of reversing a singly linked list. To take an example, given the linked list below,



The reversed linked list should look like



Which one of the following statements is TRUE about the time complexity of algorithms that solve the above problem in $O(1)$ space?

- ☒ A. The best algorithm for the problem takes $\Theta(n)$ time in the worst case.
- B. The best algorithm for the problem takes $\Theta(n \log n)$ time in the worst case.
- C. The best algorithm for the problem takes $\Theta(n^2)$ time in the worst case.
- D. It is not possible to reverse a singly linked list in $O(1)$ space.



Topic : PYQ Practice – 1 – Homework Question



The minimum number of comparisons required to determine if an integer appears more than $n/2$ times in a sorted array of n integers is

A. $\Theta(n)$

binary search

$O(\log n)$

5 4 3 (2) 2 2 2

☒ B. $\Theta(\log n)$

C. $\Theta(n \cdot \log n)$

$\frac{n}{2}$ index

$\frac{n}{2} + 1$ index

D. $\Theta(1)$

2 3 4 5 5 5 5

2 2 2 2 3 4 5

GATE 2008



Which one of the following hash functions on integers will distribute keys most uniformly over 10 buckets numbered 0 to 9 for i ranging from 0 to 2020?

- (A) $h(i) = i^2 \bmod 10$
- ☒ (B) $h(i) = i^3 \bmod 10$
- (C) $h(i) = (11 * i^2) \bmod 10$
- (D) $h(i) = (12 * i) \bmod 10$



Consider a hash table of size seven, with starting index zero, and a hash function $(3x + 4) \bmod 7$. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 3, 8, 10 is inserted into the table using closed hashing? Note that '_' denotes an empty location in the table.

- (A) 8, _, _, _, _, _ 10
- (B) 1, 8, 10, _, _, _ 3
- (C) 1, _, _, _, _, _ 3
- (D) 1, 10, 8, _, _, _ 3



Given the following input (4322, 1334, 1471, 9679, 1989, 6171, 6173, 4199) and the hash function $x \bmod 10$, which of the following statements are true?

1. 9679, 1989, 4199 hash to the same value
2. 1471, 6171 hash to the same value
3. All elements hash to the same value
4. Each element hashes to a different value

(A) 1 only

(B) 2 only

(C) 1 and 2 only

(D) 3 or 4



A hash table of length 10 uses open addressing with hash function $h(k) = k \bmod 10$, and linear probing. After inserting 6 values into an empty hash table, the table is as shown below.

Which one of the following choices gives a possible order in which the key values could have been inserted in the table?

- ~~(A)~~ 46, 42, 34, 52, 23, 33
- ~~(B)~~ 34, 42, 23, 52, 33, 46
- ☒ (C) 46, 34, 42, 23, 52, 33
- ~~(D)~~ 42, 46, 33, 23, 34, 52

A)

0	
1	
2	42
3	52
4	34
5	23
6	46
7	33
8	
9	

B)

0	
1	
2	42
3	23
4	34
5	52
6	33
7	46
8	
9	

☒ C)

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	

D)

0	
1	
2	42
3	33
4	23
5	34
6	46
7	52
8	
9	

0	
1	
2	42
3	23
4	34
5	52
6	46
7	33
8	
9	



Consider a hash table of size 11 that uses open addressing with linear probing. Let $h(k) = k \bmod 11$ be the hash function used. A sequence of records with keys

43 36 92 87 11 4 71 13 14

is inserted into an initially empty hash table, the bins of which are indexed from zero to ten. What is the index of the bin into which the last record is inserted?

- (A) 2
- (B) 4
- (C) 6
- (D) 7

$$h(43) = 43 \div 11 = 10$$

$$h(36) = 36 \div 11 = 3$$

0	87
1	11
2	13
3	36
4	92
5	4
6	71
7	14
8	
9	
10	43



Consider a hash table with 100 slots. Collisions are resolved using chaining. Assuming simple uniform hashing, what is the probability that the first 4 slots are unfilled after the first 4 insertions?

- (A) $(96 \times 96 \times 96 \times 96)/100^4$
(B) $(99 \times 98 \times 97 \times 96)/100^4$
(C) $(97 \times 96 \times 95 \times 94)/100^4$
(D) $(97 \times 96 \times 95 \times 94)/(4! \times 100^4)$

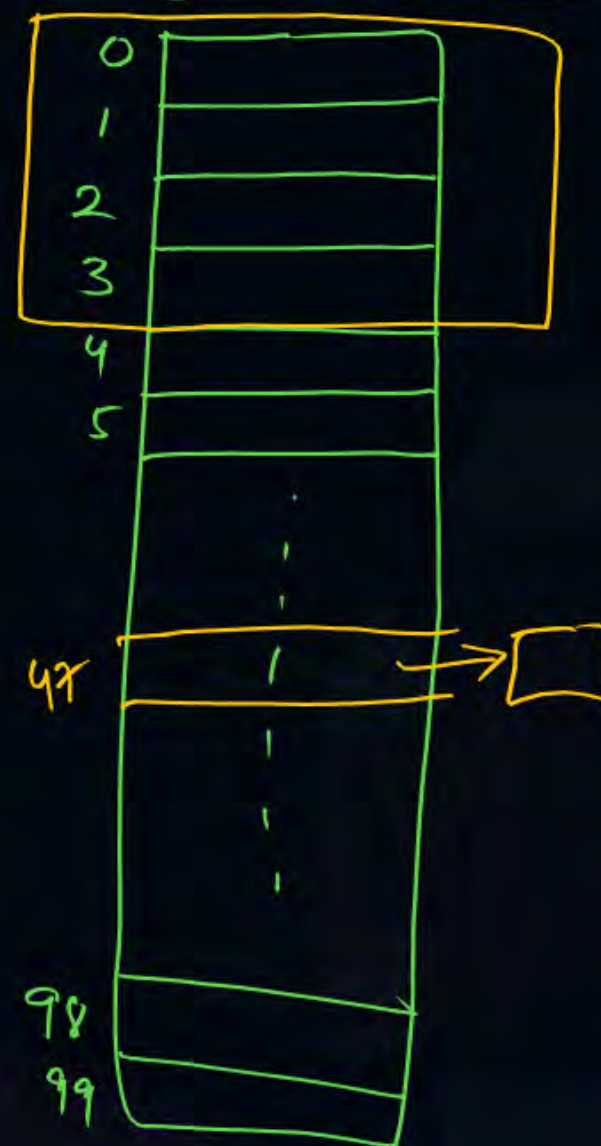
Chaining

1st 2nd 3rd 4th

$$\frac{96}{100} * \frac{96}{100} * \frac{96}{100} * \frac{96}{100}$$

If open addressing

$$\frac{96}{100} * \frac{95}{100} * \frac{94}{100} * \frac{93}{100}$$





Consider a binary tree T in which every node has either zero or two children. Let $n > 0$ be the number of nodes in T . Which ONE of the following is the number of nodes in T that have exactly two children?

- A. $n-2/2$
- ☒ B. $n-1/2$
- C. $n/2$
- D. $n+1/2$

' x ' (leaf nodes, External), No. of nodes ^{with} exactly 2 children = $x-1$ [Internal]

$$\text{Total nodes} = x + (x-1) = 2x-1$$

$$2x-1 = n$$

$$2x = n+1$$

$$x = \frac{n+1}{2}$$

$$\begin{aligned} x-1 &= \frac{n+1}{2} - 1 \\ &= \frac{n+1-2}{2} \\ &= \frac{n-1}{2} \end{aligned}$$



Topic : PYQ Practice – 2 – GATE 2023



Consider the function foo and the binary tree shown.

class Node:

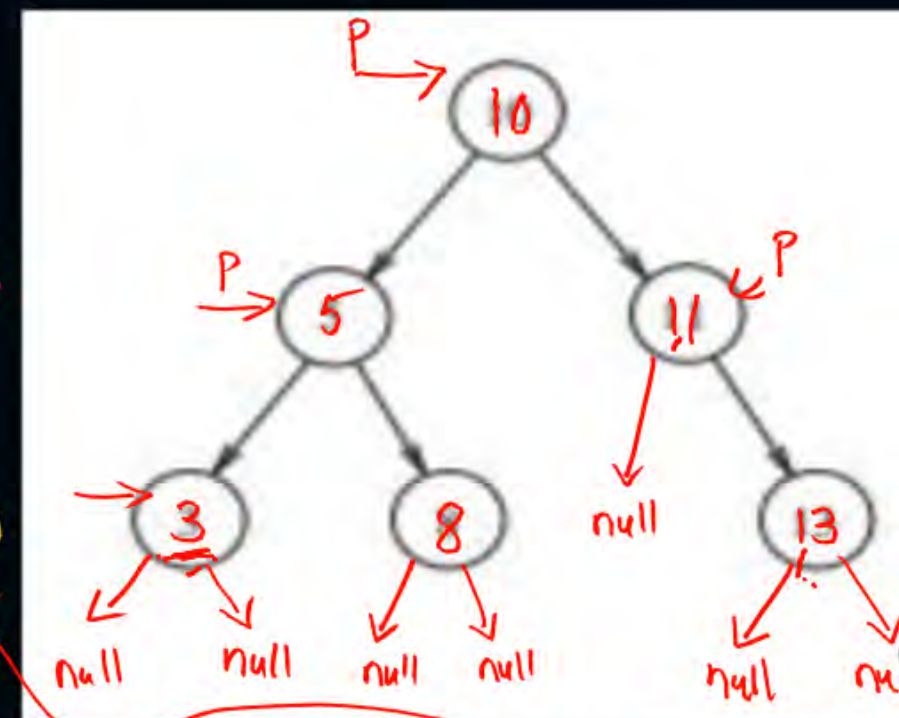
```
def __init__(self, val=0, left=None, right=None):  
    self.val = val  
    self.left = left  
    self.right = right
```

```
def foo(p):  
    if p is None:  
        return 0  
    else:  
        retval = p.val + foo(p.left) + foo(p.right)  
        print(retval, end=' ')  
        return retval
```

O/p: 3 8 16 13 24 50

struct Node

```
struct Node *left = NULL;  
int val;  
struct Node *right = NULL;
```



Print 50
 $retval = 10 + foo(p.left) + foo(p.right)$

Print 16 and return 16
 $x = 5 + f(L) + f(R)$
 $x = 3 + f(L) + f(R)$
Print 3

Print 24, return 24
 $11 + f(L) + f(R)$
 $0 + x = 13 + f(L) + f(R)$
Print 13 and return 13
 $x = 8 + f(L) + f(R)$
 $x = 8$
Print 8



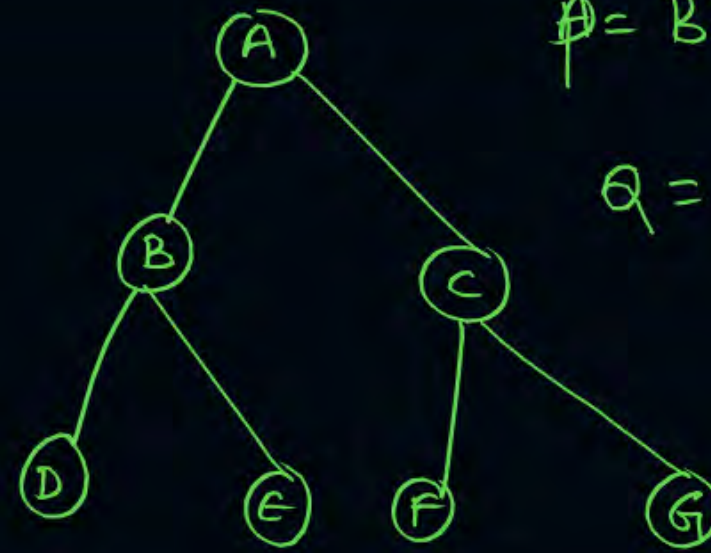
When foo is called with a pointer to the root node of the given binary tree, what will it print?

- A. 3 8 5 13 11 10
- B. 3 5 8 10 11 13
- ☒ C. 3 8 16 13 24 50
- D. 3 16 8 50 24 13



Consider a complete binary tree with 7 nodes. Let P denote the set of first 3 elements obtained by performing Breadth-First Search (BFS) starting from the root. Let Q denote the set of first 3 elements obtained by performing Depth-First Search (DFS) starting from the root.

The value of $|P-Q|$ is _____



$$P = \text{BFS} = \{A, B, C\}$$

$$Q = \{A, B, D\} \quad \text{or} \quad \{A, B, E\}$$

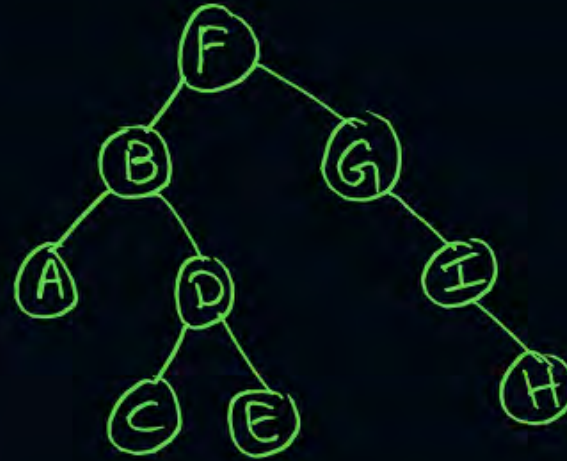
$$\begin{aligned} |P-Q_1| &= \{C\} = 1 \\ |P-Q_2| &= \{C\} = 1 \\ Q_1-P &= \{D\} = 1 \\ Q_2-P &= \{E\} = 1 \end{aligned}$$



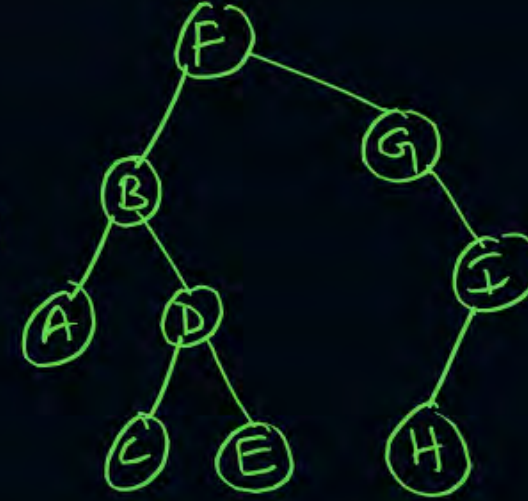
The post-order traversal of binary tree is ACEDBHIGF. The pre-order traversal is

- ~~A. ABCDEFGHI~~
- B. FBADCEGIH
- C. FABCDEGHI
- ~~D. ABDCEFGIH~~

(B)



(A)





Let T be a full binary tree with 8 leaves. (A full binary tree has every level full.) Suppose two leaves a and b of T are chosen uniformly and independently at random. The expected value of the distance between a and b in T (i.e., the number of edges in the unique path between a and b) is (rounded off to 2 decimal places) 4.25.

a and b can be same node $\Rightarrow$ distance = 0 = 8
 [a left, b Right] [a Right, b left]

With distance 2 Total 4 Possibilities + 4 Possibilities

With distance 4 Total 8 Possibilities + 8 Possibilities

With distance 6 Total 16 Possibilities + 16 Possibilities

Total Possibilities = 64



$$0 \times \frac{8}{64} + 2 \times \frac{8}{64} + 4 \times \frac{16}{64} + \frac{3}{8} \times \frac{32}{64}$$

$$= 0 + 0.25 + 1 + 3 = \underline{\underline{4.25}}$$



H/w

Consider a binary tree T that has 200 leaf nodes. Then, the number of nodes in T that have exactly two children are 199.



h/w

The height of a tree is the length of the longest root-to-leaf path in it. The maximum and minimum number of nodes in a binary tree of height 5 are

- ✓ A. 63 and 6, respectively
- B. 64 and 5, respectively
- C. 32 and 6, respectively
- D. 31 and 5, respectively



THANK - YOU